

Technology Stack

Date	27 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Technology Stack:

The **Emotion Detection & Learning Support Engine** uses a combination of frontend, backend, machine learning, NLP, AI, and development tools to provide accurate emotion detection and personalized learning support. The selected technology stack ensures scalability, maintainability, efficient data processing, and an interactive user experience.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit, HTML, CSS	Provides an interactive and user-friendly web interface for entering text and viewing emotion detection results.
2	Backend / Server-Side	Python	Handles text preprocessing, emotion detection, recommendation generation, and application logic.

3	Database / Data Storage	CSV Dataset	Stores emotion datasets and learning resources used for model training and recommendations.
4	Cloud / Hosting / Deployment	Local Host / Streamlit Cloud	Used to run the application locally and supports future cloud deployment.
5	Version Control & CI/CD	Git, GitHub	Enables source code management, version tracking, and team collaboration.
6	Third-Party APIs / Other Tools	Hugging Face Transformers (BERT), PyTorch, Pandas, NumPy, NLTK, Google Gemini API	Used for text preprocessing, emotion detection, deep learning, data processing, and AI-powered learning support.